

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large
Loads to the Interstate
Transmission System

)

Docket No. RM26-4-000

COMMENTS OF THE ENERGY TRADING INSTITUTE

Pursuant to the Notice Inviting Comments published by the Federal Energy Regulatory Commission (“FERC” or “Commission”) for the above captioned proceeding, the Energy Trading Institute (“ETI”)¹ hereby submits comments with respect to the letter by the Secretary of Energy issuing an Advanced Notice of Proposed Rulemaking regarding the interconnection of large loads to the interstate transmission system.²

ETI appreciates the opportunity to participate in this proceeding and commends the DOE and the Commission in this effort seeking to improve open and non-discriminatory access to the interstate electric transmission system by setting forth key principles of interconnection reform. Missing from these principles is a critical core principle that the Commission and the DOE should consider to efficiently shore up large load interconnection: price formation in the Regional Transmission Organization (“RTO”) and Independent System Operator (“ISO”) markets. As with

¹ ETI represents energy market participants, all with substantial interests in the wholesale electricity markets. ETI members provide important services to a wide variety of participants in such markets. Members act as intermediaries between producers and consumers of electric energy that have mismatched quantity, timing, and contract type needs. In addition, they provide liquidity by engaging in energy-related commercial transactions with a variety of market entities including, but not limited to, generation owners, project developers, load-serving entities, and investors. ETI members advocate for markets that are open, transparent, competitive, and fair—all necessary attributes for markets to ultimately benefit electricity consumers. ETI and its members have also been, and remain, active in many of the Commission’s ongoing proceedings across the various RTOs and ISOs.

² *Interconnection of Large Loads to the Interstate Transmission System*, Notice Inviting Comments, Docket No. EM26-4-000 (Oct. 27, 2015) (noticing the Secretary’s letter and the ANOPR proposed by the Department of Energy (“DOE”)) (“ANOPR”).

all investment considerations concerning the U.S. bulk electric system – whether transmission, generation, load, or otherwise – ETI urges careful consideration of price formation to achieve the policy goals contemplated in this proceeding. Market price systems can serve as “useful means of guiding overall resource allocation” for interconnection investments for large loads or otherwise.³ Indeed, the Commission has recognized that without clear pricing mechanisms, market signals and efficient system utilization are undermined, while investment signals become muted.⁴

Accordingly, in order to seek efficient interconnection investments, and in addition to the fourteen principles set forth by the DOE’s ANOPR, price formation should be included as one of the core principles. Without proper price formation, market participants and investors are left “groping about in the dark”⁵ as to what types of interconnection projects would be useful and profitable, regardless of whether and how the jurisdictional considerations are ultimately resolved.

Accordingly, in the comments below, ETI recommends that in tandem with this proceeding and other DOE initiatives, the Commission should reinvigorate its price formation agenda.

I. Communications and Correspondence

ETI requests that the following names be placed on the service list for this proceeding, and that all correspondence and communication with respect to this proceeding be addressed to the following:⁶

³ See Randal K. Quarels, Vice Chair for Supervision, Board of Governors of the Federal Reserve System, “Friedrich Hayek and the Price System” (Nov. 1, 2019) (citing to F.A. Hayek, “The Use of Knowledge in Society,” THE AMERICAN ECONOMIC REVIEW (Sep. 1945)), found at: <https://www.federalreserve.gov/newsevents/speech/quarles20191101a.htm>

⁴ *Price Formation in Energy and Ancillary Services Markets Operated by Regional Transmission Organizations and Independent System Operators*, 153 FERC ¶ 61,221, at P. 66 (2015) (“2015 Price Formation Order”).

⁵ See Ludwig von Mises, *Human Action: A Treatise on Economics*, LIBERTY FUND (ed. B. Graves 2007) (discussing the impossibility of economic calculation under socialism, whereby a centrally “planned economy is no economy at all ... [i]t is just a system of groping about in the dark.”).

⁶ Persons denoted with an asterisk (*) are those designated for service pursuant to 18 C.F.R. § 385.2010.

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II. COMMENTS

Below, ETI (A) explains why price formation efforts should be renewed in earnest, particularly in light of poor market outcomes arising from out-of-market interventions such as often-occurring unintended suppression of wholesale prices from demand response activations; (B) discusses the need for clear market signals to incentivize investments needed to address load growth; (C) discusses how competitive market prices can help improve reliability and affordability; and (D) explores recommendations for next steps.

A. The Case for Renewed Price Formation Efforts

Since 2014, the Commission has had a number of proceedings to address price formation issues in the RTO-ISO markets. Despite these efforts, price suppressing events remain a persistent problem in these markets—even as recently as with the PJM uplift that arose out of the January 2025 winter storm in the Northeast—thus undermining the signals that the prices should convey. In addition, the summer of 2025 saw increasing reliance on demand response initiatives in PJM Interconnection L.L.C. (“PJM”), which while beneficial for reliability and efficient resource allocation in the short run, have had the unintended consequence of suppressing wholesale market prices; and moreover, overuse of these programs in the presence of relatively normal spot market

prices will lead demand response providers to unsubscribe, defeating the programs' reliability enhancing purpose. These price formation problems impact all manner of electrical infrastructure investment, including interconnection with transmission, and it is important for the Commission and the DOE to renew and reinvigorate the efforts to reform price formation.

1. History of FERC's Price Formation Efforts

Starting in 2014, the Commission recognized the need for the reform of energy market pricing for the various jurisdictional RTO-ISOs. While these have been generally positive developments, the need for further reform of the RTO-ISO pricing mechanisms remains, and should be renewed in light of continual pricing distortions which will hamper efforts to efficiently allocate transmission interconnections.

In 2014, the Commission began to recognize issues related to the formation of market prices for energy and ancillary services for the various jurisdictional Regional Transmission Organizations ("RTO") and Independent System Operators ("ISO"), and noted that such prices were not reflecting the marginal cost of production, actual operating costs, or operating limits. Accordingly, the Commission was concerned with the out-of-market actions (*e.g.*, uplift payments, operator actions to secure reliability) that had been used by the RTO-ISOs that were driving such a divergence of energy and ancillary service prices from market fundamentals.⁷ Accordingly, the Commission ordered the RTO-ISOs to file reports addressing five areas of price formation: (1) pricing of fast-start resources; (2) commitments to manage multiple contingencies; (3) look-ahead modeling; (4) uplift allocation; and (5) transparency.⁸

⁷ *Price Formation in Energy and Ancillary Services Markets Operated by Regional Transmission Organizations and Independent System Operators*, Notice, FERC Docket No. AD14-14-000 (Jun. 19, 2014).

⁸ 2015 Price Formation Order at P 1.

By 2016, the Commission recognized the need for pricing reform since the RTO-ISO energy pricing mechanisms were failing “to reflect the value of providing a given service, thereby distorting price signals and failing to provide appropriate signals for resources to respond to the actual operating needs of the market.”⁹ To address these distortions, the Commission issued Order 825. RTO-ISOs were ordered to settle energy transactions in relation to the relevant timing for when energy is dispatched and intertie transactions are scheduled.¹⁰ Further, the Commission ordered the RTO-ISOs to trigger shortage pricing for intervals whereby a shortage of energy or operating reserves has been indicated for the pricing of a resource during a given interval.¹¹ When such shortages are resolved, not by a supply resource, but by the activation of retail or LDC-level demand response, it is no less critical that shortage pricing be triggered in order for wholesale prices to reflect system conditions.

In 2017, the Commission further recognized that RTO-ISO “uplift” payments to generating resources can undermine price signals leading to unjust and unreasonable rates, and that there was an overall lack of transparency associated with uplift allocation and payment.¹² As such, with Order 844, the Commission ordered the RTO-ISOs to improve uplift transparency with reports to address uplift by specific resources and zones, among other reports.¹³ For entities that serve load, uplift payments are particularly pernicious as these entities rely on forward power purchases to

⁹ *Settlement Intervals and Shortage Pricing in Markets Operated by Regional Transmission Organizations and Independent System Operators*, Order No. 825, 154 FERC ¶ 61,276, at P 2 (2016) (“Order 825”).

¹⁰ *Id.* at P. 53, 88.

¹¹ *Id.* at P. 162.

¹² *Uplift Cost Allocation and Transparency in Markets Operated by Regional Transmission Organizations and Independent System Operators*, Order No. 844, 163 FERC ¶ 61,041, at P. 21 (2018) (“Order 844”). *See also*, FEDERAL ENERGY REGULATORY COMMISSION, “Staff Analysis of Uplift in RTO and ISO Markets,” (Aug. 2014) (explaining that “Uplift credits are payments made to resources whose commitment and dispatch by an RTO or ISO result in a shortfall between the resource’s offer and the revenue earned through market clearing prices”), found at: https://www.ferc.gov/sites/default/files/2020-05/08-13-14-uplift_3.pdf

¹³ Order 844 at P. 27, 50, 74.

hedge their price risk exposure. Uplift charges represent an uncertain and unhedgable cost that adds to the risk premium that the competitive market requires for being in the business of serving full-requirements customers, and therefore raise costs to end consumers.

2. ISO-RTO Prices Remain Subject to Out-of-Market Distortions

Other than addressing the various RTO-ISO compliance filings responding to these price formation orders, the Commission's price formation efforts have stalled, and material issues remain, including situations that penalize behavior that could improve coherent price formation. These material issues include: (a) persistent use of uplift allocations and other after-market adjustments; (b) wholesale market price distortions from the lack of demand response transparency; and (c) underfunding of congestion derivative instruments.

Each of these out-of-market situations have led to poor price formation, thereby resulting in inefficient dispatch and investment signals, further increased reliance on non-market solutions, and reduced transparency and investor confidence. Accordingly, the DOE should call on FERC to reopen and update the price formation proceedings, incorporating lessons from recent market events.

a. Uplift Allocation and Other After-Market Adjustments Remain a Problem

Most recently with the January 2025 winter storm, PJM assessed uplift costs to market participants, including with out-of-market make-whole payments and lost opportunity cost credits.

As ETI explained:

“[R]ecent extreme winter weather has once again highlighted critical flaws in PJM's market structure, resulting in over \$400 million in uplift costs in 2025 year-to-date—expenses that cannot be hedged by load-serving entities through forward energy purchases and which are often passed directly to consumers. These out-of-market costs signal a fundamental flaw in how the market is designed; a problem that has been known to PJM and FERC for some time. It is imperative that PJM, PJM stakeholders, FERC and the

current administration prioritize reforms that ensure these costs are reflected in energy prices rather than hidden in unhedgable charges.”¹⁴

This sort of price suppression during a scarcity event prompted by a winter storm makes it difficult for investors to reasonably assess the true economic value and feasibility of electricity projects.

Similar investment uncertainty arose when a vast amount of generation did not perform for the PJM capacity market during Winter Storm Elliott in December 2022. Under the PJM tariff, penalties were assessed for allegedly non-performing resources, while bonuses were due to over-performing resources. Ultimately, FERC mediated a settlement agreement between the various generators and PJM, whereby PJM agreed to reduce the assessed penalty amount by 31.7%, which also reduced the bonus by the same amount.¹⁵ Such after-the-fact adjustments make it difficult for investors to assess the potential viability of electricity projects.

b. Demand Response Programs Not Factored into Wholesale Market Price Formation

State initiated demand response programs can be efficient mechanisms for operational flexibility and allocation of electricity usage within a region.¹⁶ The problem is that some demand response programs occur at the local distribution level and are not integrated into the RTO-ISO price formation process.¹⁷ Since these demand response programs are not integrated into the wholesale price formation process, the RTO-ISOs do not have visibility into the demand (load) that takes place in these areas. For example, a 2022 study estimates that approximately 5% of peak load from the Baltimore Gas & Electric region was not accounted for in the load estimates for the

¹⁴ ENERGY TRADING INSTITUTE, Press Release, “Winter Weather Highlights Market Design Flaws, Underscoring the Need for Reliable and Affordable Energy Solutions” (published Mar. 18, 2025).

¹⁵ *PJM Interconnection L.L.C., et al.*, 185 FERC ¶ 61,204, at P 37 (2023).

¹⁶ *Modernizing Electricity Market Design*, FERC Docket No. AD21-10-000, “Post-Technical Conference Comments of Appian Way Energy Partners LLC Regarding the Need for Better Integration of Demand Response into Wholesale Market Price Formation,” at p. 2 (Feb. 4, 2022) (“Appian DR Comments”).

¹⁷ *Id.* at p. 3.

corresponding PJM market.¹⁸ RTO-ISOs that fail to account for the impacts of demand response programs have depressed wholesale prices in such locations.¹⁹

These sorts of poor market outcomes, deriving from the lack of transparency of State demand response programs to the RTO-ISO markets, harm electrical infrastructure investment. It will be important for the RTO-ISOs to have a clear understanding of the demand profiles that derive from both traditional loads and newer large load resources so that proper wholesale price formation and corresponding investment can be achieved. Notably, these issues are salient in Texas, where under Senate Bill 6 (SB6), the Public Utility Commission of Texas (“PUCT”) is required to develop large-load demand management programs to improve reliability during times of high demand on the grid.²⁰ Under the mandatory program, new large loads can be curtailed under emergency conditions,²¹ with such conditions triggering scarcity pricing in ERCOT.²² Triggering scarcity pricing when scarcity conditions dictate demand response seems like a clear best-practice in price formation. Without explicit FERC direction, ISOs will simply continue to see state-sponsored demand response initiatives as missing load that suppresses prices even in circumstances where true system conditions dictate high spot prices, or even scarcity.

c. Underfunding of Congestion Derivatives Contracts

Currently, in the California Independent System Operator (“CAISO”) market, price formation in the spot and forward energy markets is hampered by chronic underfunding of the

¹⁸ *Id.* at p. 3-4 (explaining study performed by Appian Way on local demand response program impacts on ISO-RTO price formation).

¹⁹ *Id.* at p. 4.

²⁰ *An Act relating to the establishment of the Texas Energy Insurance Program and other funding mechanisms to support the construction and operation of electric generating facilities*, S.B. No. 6 (eff. Jun. 21, 2025) (“Senate Bill 6”), found at: <https://legiscan.com/TX/text/SB6/id/3248460>

²¹ *Id.* at Sec. 39.170.

²² See Electric Reliability Council of Texas, Nodal Protocols, Section 2: Definitions and Acronyms (Sep. 1, 2025) (defining “Operating Reserve Demand Curve” as a scarcity pricing mechanism).

Congestion Revenue Rights (“CRR”) instruments. Since CAISO instituted reforms to its CRR allocation process in 2019, CAISO has been aware of extreme levels of underfunding allocated to owners of CRRs,²³ with underfunding levels approaching and often exceeding 100%. A prominent example arose in February 2022, when a constraint along the Moraga-Claremont 115 kV line and other lines were substantially underfunded because the offset revenue was higher than the notional revenue of the constraint for 89 percent of the hours between January 31 and March 31, 2022.²⁴ As the CAISO explained, “[c]onsequently, scheduling coordinators holding CRRs on this constraint needed to pay for holding CRRs settled on the constraint.”²⁵ As the CAISO reported on its website, an underfunding of \$7.92 million arose in February 2022 associated with the constraint 33020_MORAGA_115_32780_CLARMNT_115_BR_2_1, for which CRR holders had to pay even though they should have received payment based on LMP pricing and the related congestion.²⁶

²³ See California Independent System Operator Corporation, “CRR Market Analysis Report,” at p. 42-64 (May 12, 2020), found at: https://stakeholdercenter.caiso.com/initiateddocuments/congestionrevenueightsmarketanalysisreport-may12-2020.pdf?utm_source=chatgpt.com

²⁴ See *Calif. Indep. Sys. Operator Corp.*, Market Parameters Enhancements, Attachment C – Final Proposal, FERC Docket No. ER23-2020-000, at p. 9 (May 31, 2023).

²⁵ *Id.*

²⁶ See *Calif. Indep. Sys. Operator Corp.*, Congestion Revenue Rights (Feb. 2022) (“CAISO Feb-2022 Report”), found at: <http://www.caiso.com/MonthlyMarketPerformanceReports/feb-2022/congestion-revenue-rights.html>

A summary of 2024 and 2025 CAISO CRR underfunding is in the table below. In 2024, CAISO experienced 44 constraints with underfunding exceeding 100%, with a total of \$75 MM in underfunding; 70 constraints with underfunding exceeding 75%, with a total of \$96 MM in underfunding; 100 constraints with underfunding exceeding 50%, with a total of \$113 MM in underfunding; and 151 constraints with underfunding exceeding 25%, with a total of \$235 MM in underfunding. In 2025, the situation improved

CAISO CRR Underfunding - 2024				
Underfunding (%)	# of constraints	Notional Revenue	Under Funding	Average Underfunding
>100	44	\$ 54,210,033	\$ (75,006,395)	138%
>75	70	\$ 74,809,723	\$ (92,520,503)	124%
>50	100	\$ 112,622,535	\$ (113,072,575)	100%
>25	151	\$ 436,162,581	\$ (235,463,131)	54%
Total Including all constraints	426	\$ 679,975,665	\$ (236,654,449)	35%

CAISO CRR Underfunding - 2025 (through 11/19/25)				
Underfunding (%)	# of constraints	Notional Revenue	Under Funding	Average Underfunding
>100	31	\$ 6,758,381	\$ (9,187,352)	136%
>75	55	\$ 24,525,716	\$ (24,378,842)	99%
>50	80	\$ 140,838,551	\$ (90,528,333)	64%
>25	133	\$ 436,835,985	\$ (194,072,542)	44%
Total Including all constraints	448	\$ 633,050,635	\$ (196,907,203)	31%

somewhat, with much of the underfunding problem shifting into the bucket with 50-75% underfunding. In 2025, CAISO experienced 31 constraints with underfunding exceeding 100%, with a total of \$9 MM in underfunding; 55 constraints with underfunding exceeding 75%, with a total of \$24 MM in underfunding; 80 constraints with underfunding exceeding 50%, with a total of \$91 MM in underfunding; and 133 constraints with underfunding exceeding 25%, with a total of \$194 MM in underfunding.

Commission precedent recognizes that underfunding of congestion derivatives is a severe problem and clearly supports allocation of costs based on cost-causation principles. FERC precedent regarding cost allocation is clear: assign costs specifically when specific beneficiaries can be identified and assign costs broadly when this is not possible. The Commission has held that underfunding harms CRR holders and harms transmission customers by devaluing the congestion rights that are allocated to them.²⁷

²⁷ *PJM Interconnection, L.L.C.*, 158 FERC ¶ 61,093 at P 78 (2017).

Yet, CAISO CRR underfunding persists, fundamentally, from non-cost-causation sources, such as entities that use CAISO's grid not paying for the congestion they cause, leaving CRR holders to foot the bill for unpaid congestion.²⁸ Some underfunding also results from modeling differences between the CRR market (forward term) and the CAISO integrated forward market (spot, next day).²⁹ CAISO has had multiple proceedings to attempt to resolve underfunding issues over the past five years but has not made significant progress³⁰, perhaps due to the mistaken view that since load is no longer allocated the underfunding charges directly given CAISO 2019 CRR reforms, load is not ultimately harmed by the underfunding. Moreover, underfunding will certainly be exacerbated by the CAISO Extended Day-Ahead Market ("EDAM") changes as currently proposed.³¹ Further, the Commission Staff acknowledges that complex seams issues are arising in the Western Interconnection with the development of EDAM and SPP Markets+. ³² As a result

²⁸ For instance, CAISO in 2023 filed to eliminate the shift factor cutoff threshold that had been causing suppression of load-zone prices to ignore the congestion the load zone causes on certain constraints, with the associated underfunding transferred to CRR contracts with exposure to those constraints. *See California Independent System Operator Corporation*, "Market Parameter Enhancements," FERC Docket No. ER23-2020-000 (May 31, 2023), *approved in part*, 184 FERC ¶ 61,119 (2023). CAISO has identified that allocation of underfunding persists in the shift factor cutoff for imports and large generators, and well as for loop flows on CAISO's network, both of which represent non-cost-causation-based allocations of underfunding. Kyle Pelham, et al., CALIFORNIA INDEPENDENT SYSTEM OPERATOR CORPORATION, CAISO CRR Forum, "Analysis of CRR Market Performance," at p. 91 (Feb. 27, 2025), found at: <https://stakeholdercenter.caiso.com/InitiativeDocuments/Presentation-Congestion-Revenue-Rights-Enhancements-Feb-27-2025.pdf>

²⁹ Kallie Wells, WESTERN POWER TRADING FORUM, CAISO CRR Forum, "CRR Enhancements, Pathway Forward," (Sep. 2025), found at: <https://stakeholdercenter.caiso.com/InitiativeDocuments/WPTF-Presentation-Congestion-Revenue-Rights-Enhancements-Sep-08-2025.pdf>

³⁰ *See e.g., California Independent System Operator Corporation*, "Initiative: Congestion revenue rights enhancements," CAISO CRR Forum (eff. Sep. 26, 2024), found at: <https://stakeholdercenter.caiso.com/StakeholderInitiatives/Congestion-revenue-rights-enhancements>

³¹ CAISO's EDAM proposal will exacerbate underfunding in two ways. First, CAISO has amended its EDAM tariff to provide non-CAISO firm physical rights holders (NITS and point-to-point) with a perfect hedge for their flow on CAISO constraints, even in circumstances with transmission outages reduce the transfer capacity of CAISO's network (i.e. rather than pro-rating the reduction in transfer capacity as would normally happen with existing physical rights, CAISO will fully fund non-CAISO firm rights, leading CRR holders to foot the bill). And second, while non-CAISO holders of firm rights are provided full-funding on CAISO constraints, CAISO firm rights holders are explicitly provided zero funding on non-CAISO constraints.

³² Federal Energy Regulatory Commission, Staff Whitepaper, "Seams Coordination in the Western Interconnection," at p. 1-5 (Nov. 6, 2025), found at: <https://primarynewssource.org/wp-content/uploads/Seams-White-Paper-2025-11.pdf>

“[u]nderfunding uncertainty increases risk and lowers the expected CRR value in the auction, depressing CRR bid prices and reducing the value of ARR.”³³

It is important for CRRs to be funded appropriately as they are critical to forming forward prices for signals to future investment. Developers are using CRRs and the instruments need to function and price appropriately to for developers to properly hedge physical exposure.³⁴ As such, congestion derivative underfunding should be addressed as part of price formation reforms.

B. Load Growth and the Need for Market-Based Investment Signals

DOE’s projections show record levels of demand growth from data centers, artificial intelligence computing, and manufacturing.³⁵ Meeting this growing load will require significant new generation, transmission, and flexible resources—all of which depend on clear and durable RTO-ISO price signals. Indeed, competitive markets have proven to be able to facilitate cleaner, more reliable power at a more cost effective price for consumers.³⁶ As demonstrated above, RTO/ISO market designs that fail to properly reflect scarcity, congestion, and risk appropriately, have the unfortunate consequence of deterring capital investment.

³³ Abram W. Klein, APPIAN WAY ENERGY PARTNERS, CAISO CRR Forum, “CRR Market Design: How CRRs Benefit End-use Customers,” (Jan. 28, 2025), found at: <https://stakeholdercenter.caiso.com/InitiativeDocuments/Presentation-Appian-Way-Congestion-Revenue-Rights-Enhancements-Jan-28-2025.pdf>

³⁴ See Cathleen Colbert, VISTRA CORP., CAISO CRR Forum, “Congestion Revenue Rights Enhancements Working Group,” at p. 9 (Jan. 28, 2025) (requesting that CAISO expand its analysis to “better inform ‘key drivers’ of issues that may be undermining ability of day-ahead market to collect as close to accurate congestion rents as possible to improve funding to the congestion revenue rights holders”), found at: <https://stakeholdercenter.caiso.com/InitiativeDocuments/Presentation-Vistra-Congestion-Revenue-Rights-Enhancements-Jan-28-2025.pdf>; see also, Joshua Reichert, MORGAN STANLEY, CAISO CRR Forum, “Congestion Revenue Rights Enhancements working group call,” at p. 5 (Jan. 28, 2025) (suggesting that CAISO should consider better incentives for accurate outage forecasting as a mitigant to underfunding), found at: <https://stakeholdercenter.caiso.com/InitiativeDocuments/Presentation-Morgan-Stanley-Congestion-Revenue-Rights-Enhancements-Jan-28-2025.pdf>

³⁵ Department of Energy, “2024 Report on U.S. Data Center Energy Use,” (Dec. 20, 2024) (projecting US electricity demand to double or triple by 2028 owing to data center load growth, among other factors), found at: <https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers>

³⁶ Electric Power Supply Association, “Getting to the Truth on Competitive Electricity Markets” (Feb. 2, 2023) <https://epsa.org/getting-to-the-truth-on-competitive-electricity-markets/>

Accordingly, ETI recommends that the DOE and the Commission work with the RTO-ISOs to strengthen locational pricing mechanisms to more properly reflect system conditions. Current prices of forward markets tell market participants when to store and/or consume commodities, and help investors determine the relative values of time and place for the feasibility, location, and development of infrastructure projects. Accordingly, promoting forward markets that provide transparent price signals facilitates generation and transmission investment. An independent analysis by London Economics International (“LEI”) estimated that forward price markets as derived from congestion derivatives like FTRs save consumers in a market like PJM up to \$1.2 billion annually by enhancing liquidity, transparency, and hedging in the forward market.³⁷ Another LEI analysis for the Midcontinent Independent System Operator, Inc. (“MISO”) showed that the Stage 2 Auction Revenue Rights (“ARRs”) would not be fully funded without the price liquidity provided by financial participants in that FTR market.³⁸ These findings demonstrate the value of sound forward market prices to consumers and to overall market efficiency.

It would be useful to consider expansion of congestion derivatives markets (e.g., Financial Transmission Rights, CRRs) as essential hedging tools that provide term forward price signals to underpin long-term investment. This is particularly critical as additional solar and wind generation is integrated into the grid. In many markets, investor-owned utilities and other market participants have requested congestion derivatives that distinguish between solar and non-solar time-of-use to assist in hedging physical assets.³⁹

³⁷ London Economics International LLC, *Review of PJM’s Auction Revenue Rights and Financial Transmission Rights* at 16 (Dec. 16, 2020) (“LEI PJM Report”).

³⁸ London Economics International LLC, *Independent Evaluation of MISO’s Auction Revenue Rights (“ARRs”) and Financial Transmission Rights (“FTRs”)* at 48 (Jan. 12, 2023).

³⁹ VISTRA CORP., “Granular Product Type for CRR TOU,” ERCOT Nodal Protocol Revision Request, NPRR1292 (Jul. 22, 2025) (requesting the creation of a CRR time-of-use product to align with current resource mix, including solar), found at: <https://www.ercot.com/mktrules/issues/NPRR1292>

That said, another difficulty is that while generation projects have been clearing through the interconnection queue, such as with PJM, “[n]ew supply has been slow to connect to the grid.”⁴⁰ Sometimes the pause in generation projects being fully developed has been due to State siting and permitting processes.⁴¹ Better and more reliable price formation will help investors better assess generation developments and retirements, but it will also help State regulators understand the need for further generation infrastructure development and improving their siting and permitting processes.

C. Competitive Market Prices Can Help Strengthen Reliability and Affordability of Energy

Markets remain the most effective means of balancing reliability and affordability—but only if prices reflect true system conditions. DOE and the Commission should work with the RTO-ISOs to provide for better price formation to accurately reflect system conditions and market fundamentals and facilitate markets for congestion hedging, which can reduce long-run costs by incentivizing efficient resource siting and flexible capacity.

As demonstrated above, administrative interventions such as uplift payments, which are not hedgeable through forward contracts, may seem politically convenient, but raise consumer costs directly as a pass through in some circumstances and indirectly by raising the risk premia for serving full-requirements load and deterring market-based investment. Coordinated market design improvements across RTOs/ISOs are critical to maintaining consistent reliability frameworks. There is precedent for market designs helping with structural changes in energy markets, such as

⁴⁰ Asim Haque, “State frameworks are critical to addressing PJM affordability,” UTILITY DIVE (Oct. 2, 2025).

⁴¹ *Id.* (explaining that “If energy infrastructure projects don’t receive state permitting approvals, it won’t matter how many projects PJM pushes through its generation interconnection queue.”).

with the transition from coal generation to natural gas generation on the supply side. The same could be the case for the demand side with respect to the advent of new large loads.

D. DOE and FERC Coordination and Next Steps

ETI commends the DOE for using its statutory authority to initiate this ANOPR, and ETI recommends that DOE and the Commission consider the reforming of RTO-ISO price formation to be added as a core principle to this interconnection reform.

Further, ETI recommends that the Commission reopen its price formation inquiry to enact needed RTO-ISO pricing reforms, and absent action initiated by the Commission, for the DOE to formally exercise its statutory authority requesting that the Commission to do so.⁴² In reopening this inquiry, the DOE and the Commission should convene a joint technical conference on price formation, scarcity pricing, forward curve development, and integration of AI-driven load forecasting.

Additionally, ETI recommends that the DOE use its authorities to prioritize market-based reforms as part of DOE's grid modernization and transmission funding programs. These markets have provided significant value to consumers over the past two decades and it is critical that we utilize these tools to maintain affordability as we go through this transition. These market structures are designed to shift risk away from consumers and to entities that can manage that risk. When we have non-market-based solutions, the consumer ultimately pays a higher price.

Accordingly, ETI stands ready to collaborate with DOE and FERC staff and to provide solutions to the examples of poor price formation outcomes across the RTOs/ISOs as discussed above.

⁴² 42 U.S.C. § 7173(a).

III. CONCLUSION

Reinvigorating competitive market design is the most cost-effective and scalable way to meet rising load while maintaining reliability. ETI encourages DOE to reaffirm its support for market-based policy mechanisms and to partner with the Commission and market participants to address structural pricing deficiencies that inhibit efficient investment.

ETI offers its continued engagement and expertise in ongoing DOE and FERC initiatives.

Dated: November 21, 2025

/s/ Casey Khan

Casey Khan

Terence T. Healey

Attorneys for Energy Trading Institute

CERTIFICATE OF SERVICE

I hereby certify that I have this day served the foregoing document upon each person designated on the official service list compiled by the Secretary in this proceeding.

Dated at Washington, D.C. this 21st day of November, 2025.

/s/ **Charles Beauregard**
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